

What This Book Is For

Why every chapter begins with the file rather than the question

84-355 Democracy's Data

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Plenty of books will teach you statistics. They will show you what the numbers mean once you have them, how sure you can be that a difference is real, and how a chart or a model can be built to mislead. Plenty of others will teach you the tools: how to clean up a table, join two of them together, draw the graph. Both kinds of book are good, and there are a great many of them.

Neither one is about the thing sitting between them: **the dataset itself**. Somebody made it. They had a reason, and the reason was almost never your question. Some things they wrote down because a law made them.

Other things they wrote down because they felt like it. And some things they could not have written down even if they had wanted to. What you are handed is a file that will answer a few of your questions and has already ruled out the rest.

That is where the mistakes live – not in the arithmetic, which is usually fine, and not in the code, which usually runs.

So this is a book about **where the numbers of American democracy come from**. Its subject is the record itself, in both directions: what a public is able to find out about its own democracy, and what a government has written down about the people it governs. Most of what a citizen can know about an election, a district, a neighborhood or a police force was settled long before anybody computed anything. It was settled by which agency was obliged to keep a record, what that record was kept for, and what it was permitted to ask.

You will pick up other things along the way. There is arithmetic here, and some statistics, and code behind every number on these pages. None of that is the aim. The aim is that by the end you know **what exists**. Who counts the population, who publishes the returns, who keeps the rolls, what a campaign is required to disclose, what a traffic stop leaves behind.

And that you can open one of those files and, before you compute anything at all, say four things about it.

- Who made this, and why?
- What is one row?
- If I want to compare two groups, what am I dividing by?
- Which of my questions can it answer, which can it not, and which has it quietly made unanswerable?

Knowing a source *means* being able to answer those four about it. And that is what decides whether the statistics and the programming were worth doing.

01 What one of those mistakes looks like

Two files of American presidential returns in this book, one for 2020 and one for 2024, came from the same source and were assembled by the same person. Matching them on the federal code for each county – the operation nearly every comparison of two elections rests on – returns 3,104 tidy rows. One of those rows says that between 2020 and 2024 the District of Columbia moved 0.25 points *toward* the Democrats, while the median county in the same table moved 1.53 points the other way.

Before you read on, commit to an answer. Behind that code the place had changed: one file means something by 11001 that the other does not. Write down what the match does with a pair like that. Refuse it, drop the row, print a warning, come back short of the row count you were expecting, or hand you a complete and tidy table with nothing visibly out of place. Pick exactly one.

The row is false, and it is false in a way no arithmetic will catch. Its 2020 half is the whole District of Columbia: 344,356 ballots. Its 2024 half is Ward 1: 40,463 ballots, about 12% of the city. Both files carry a row labeled 11001, which is the federal government’s code for the District. In one of them that string means the city. In the other it means an eighth of it.

Now count what did not go wrong. The match is the right operation and it was performed correctly. The arithmetic that follows it is correct. No value is missing, no field is malformed, nothing is out of range, and no statistical procedure was misapplied because none was applied. There was no warning, no error, and no shortfall in the row count to notice. Everything a careful person checks came back clean.

A statistics course would not have caught this, because nothing statistical went wrong. A coding course would not have caught it either, because the program did exactly what it was told. The result was a clean table and a believable number that happened to be false.

The one fact that would have saved you is not in a statistics book and not in a programming manual. It is that in 2024 the compiler of that file took the District’s returns from a source that reports by ward. And that a federal code names a place, rather than promising anything about what a row contains. It is a fact about the file: about who made it, from what, for what purpose, and under what obligation. That is the subject of this book.

02 Six parts, and each one is a place data comes from

The book is organised by **where the data came from**, not by what it is about. That is unusual, and it is the whole design. Turnout shows up in four different parts of this book, and it means something slightly different in each, because a different organisation produced it for a different reason. Grouping by topic would hide that. Grouping by source is what makes it visible.

Part	Chapters	What it covers
I. The Census Bureau	21	One agency that counts people and places, and where population numbers come from

Part	Chapters	What it covers
II. Surveys	17	Asking a sample of people questions: public opinion, and polls that predict elections
III. Elections	43	The certified result, and the machinery that produced it
IV. Records of Political Actors	17	What a member of Congress, a campaign or a lobbyist leaves on the record
V. Records of Ordinary People	4	Records made to run a system, about people who never volunteered for anything
VI. Putting Data Together	7	Combining the sources, which is where the hard questions and the wrong answers are

Part I is the Census Bureau, which is one agency that counts people and places, and the place nearly every population number in American politics comes from. It goes first because almost every other part borrows a number from it. If you want to know what share of a county voted, somebody has to tell you how many people live there, and it is the Bureau.

Part II is surveys — asking a sample of people questions and using their answers to describe everyone. You will already have met one in Part I, because the American Community Survey is a survey, and a compulsory one. Here you meet the voluntary kind: the long-running academic studies that measure what the public believes, and the political polls that try to say what will happen in November. This part is where the trouble with surveys gets examined rather than mentioned.

Part III is elections, and it comes in two halves that are easy to confuse. One is the **result** — who won, by how much, published by the officials who certified it. The other is the **machinery**: who was registered, who was removed from the rolls, how many ballots were rejected, what a precinct even is. Both are called “election data” and they are not the same thing, and a surprising number of published claims come from quietly swapping one for the other.

Part IV is what political actors leave behind. Every member of Congress votes on the record. Every campaign files what it raised and spent. Every lobbyist registers. None of that was collected to help you, but all of it is public, and the row is always about somebody who chose to run for office or work in politics. This is the thinnest part of the book and the one most likely to grow.

Part V is what ordinary people leave behind, which is the same kind of record with one difference that changes everything: nobody in it volunteered. A traffic stop, a jury summons, a grade stamped on a neighbourhood in 1937. These records were made to run a system, not to describe the people caught in it, and reading them as descriptions is one of the most common serious errors in this field.

Part VI puts them together. Everything before it works with one source at a time. Here you combine them. You will guess a voter's race from a name and an address, estimate how a group voted from precinct totals, and grade a district map against a rule somebody wrote down.

That is where the genuinely interesting questions live. It is also where nearly all the wrong answers come from.

03 What a chapter does, in the same order

The parts differ; the moves do not. Not every chapter makes every one, so the count beside each says how many of the 116 chapters do — measured by reading them, like everything else on this page.

The move	What that means	Chapters
Where it came from, and why	Who produced this file, under what obligation, and for whose purpose — which was almost never yours. The address is printed, so you can go and get it yourself	109
What it actually looks like	One real record, shown in full before anything is summarised — because a row is where the surprises are. The chapters that skip it have nothing small enough to print	56
What it says	Summary numbers and figures, built from the file in front of you, with the wrong reading shown as well as the right one. Every number is computed from the data as the page is built, never typed in	116
What it cannot say	The question this source will not answer however carefully you ask — a limit built into the file, not into the analysis	110
Your turn	The tables the figures rest on, linked so you can open them in a spreadsheet, and questions the chapter did not answer	110

04 Commit to an answer before you turn the page

104 of the 116 chapters stop somewhere before their central result and ask you, in so many words, to write down what you expect. You met one earlier.

This is not a quiz and nothing depends on getting it right. It is there because the interesting quantity in each chapter is not the number itself but **the distance between the number and what a reasonable person expected**. A guess you have not written down is one you will quietly revise the moment you see the answer, and you will come away believing you knew all along.

So the prompts are worth honoring, and they are worth honoring in writing. The chapters are built around the moment they turn, and the turn only lands on someone who had committed to the other side of it.

05 One real dataset, one real question, and no code to write

Each chapter takes **one real dataset and one real question**, and goes from a single record to a conclusion you could defend. It does not skip the part where the obvious answer turns out to be wrong.

The data is not simplified for teaching. Every chapter folder carries its own `data/` directory holding the actual files, and each of those directories also holds the script that built them, and beside both a stamp listing every one of those files with its size, its checksum

and the date it was last written. Where a source could not be obtained cleanly, the chapter says so rather than substituting something tidier.

You will not be asked to write any code. This is deliberate and it is not a concession. The code is written for you and it works, and every number on these pages came out of a file, by a route somebody can check. But writing it is not the exercise. The exercise is interpretation: saying what a table shows, what it does not show, and what would have to be true for its conclusion to hold. Making students author code turns a course about evidence into a course about syntax, and the people most likely to be turned away by that are exactly the people who came for the substance.

The chapters do not depend on one another and may be read in any order. They cross-reference each other often, because the same problems recur, but none is written to assume you have read another. Start with whichever question you care about.

06 Four questions, asked of every dataset in the book

This book asks each of its chapters to stop, before it argues anything, and describe the data it is about to use – answering the same four questions in the same order.

The question	What it catches
What is the source?	A file built for somebody else's purpose, and nobody's obligation to keep it right
Why is it appropriate to this question?	A unit of analysis that cannot carry the question being asked of it
What are the challenges – with the data, or with measuring the thing at all?	A quantity that was never observable, dressed as one that was
Is it clean, or does it need cleaning?	Decisions made on your behalf, upstream, that move the answer

They are not boilerplate. Each one is a distinct way a conclusion fails, and the order matters.

The source, because purpose is inherited. A file was built by somebody to do something, and that something is almost never your question. A voter registration file exists to send one person to the right ballot. The chapter on voter files finds that most of its columns are doing exactly that job and nothing else. That is why the file is excellent for running an election and poor for describing an electorate.

American election returns are stranger still. There is no national election administrator, so no agency produces them. Every county file anyone has ever used was assembled privately from fifty-one separate state records. That is not an accuracy problem, because the totals are close. It is an accountability problem, and it is the reason 11001 could change meaning with nobody to announce it.

Appropriateness, because the unit decides the question. A dataset does not answer questions. A dataset at a particular level of aggregation answers a particular kind of question, and destroys the rest. Precinct returns can tell you how a neighborhood voted and can never tell you how a person voted, because the secret ballot is the deliberate destruction of individual-level data. Push a method past that line and it does not fail loudly.

The chapter on levels of aggregation runs a technique still used in federal voting-rights litigation, and gets an estimate above 100%, with an excellent fit statistic sitting on top of it.

Challenges, because some things were never measurable. This question has two halves, and the second is harder.

Some problems belong to the file: holes, odd codes, a question a third of people skipped. Those you can usually work around.

Other problems belong to the world. The traffic-stop chapter opens on a comparison that cannot be fixed. Divide stops by the residents of San Francisco, and Hispanic drivers are stopped 0.86 times as often as white drivers — the lowest of any large group. Divide the same stops by the people who actually drive to work, and the figure is 1.01. That is essentially even.

Which bottom number is right? Neither. Nobody ever counted the people who were at risk of being stopped, so the comparison has no correct denominator. A chapter that hid that behind one confident ratio would be worse than useless.

Cleaning, because cleaning is a set of decisions. “Clean the data” sounds like tidying. It is a sequence of choices, each of which can move the answer, and most of which are invisible once made.

Read the federal county code as a number instead of as text, and 328 of 3,160 rows silently lose a digit. That is not a random 10%. It is precisely the states whose codes begin with zero, Alabama through Connecticut. They will then fail to match any other government file, without complaint.

Elsewhere the cleaning is a judgment rather than a bug. In one state’s registration file 34,522 people have a date of birth of 1 January, about 1.9 times what the calendar alone would produce. Those are placeholders, entered where the real date was unknown.

Dropping them is defensible and so is keeping them, and the two choices give different answers, because the people whose birth date was never properly recorded are not a random sample of anybody. What is not defensible is doing either one silently.

07 The same half-dozen things go wrong

Across 116 chapters and a great many unrelated sources – a Depression-era mortgage appraisal, a Senate lobbying form, a jury panel, a prediction market, a survey run since 1972 – the failures repeat. Six of them account for most of this book.

A code outlives the thing it names. 11001 is one instance. Precinct geography is worse: in a single Georgia county between 2020 and 2024 several precinct names vanished from the returns, and most were places where the polling station moved and the ground did not. One was a genuine split, and nothing in either file tells the two apart. Names and codes drift independently, so neither is a stable identifier, and a precinct that “swung twelve points” may simply have acquired three hundred houses.

A denominator nobody states. The traffic-stop ratios above are one form of it. Office-holder age is the cleanest. Whether Congress has aged faster or slower than the country it governs depends on whether you compare against life expectancy at birth or remaining life expectancy at 65. Two answers with opposite signs, one row apart in the same published table. The chapter on ranked ballots finds a winner’s “majority” is a majority of the ballots still active, not of the ballots cast.

A boundary invisible because it is the line the data was averaged on. Along Detroit's northern edge the census records a fall of 37.2 points in the Black share of the population across a single road. That road is the city limit and also the county line, so not one county file in the country contains the step. Each of the three counties has already averaged its own side of it together with a great deal of ground far away.

The line is not concealed and not disputed. It is invisible by arithmetic. Hold the people and the categories fixed, change only the size and angle of the box you average over, and the answer moves.

A probable match treated as a certain one. Somewhere in New Jersey two people who have never met share a first name, a last name and a date of birth. In a file of 6,657,135 registrants that is not a coincidence but arithmetic. And for about a decade a group of states treated matches like that as evidence of double registration.

The same mistake appears in gentler forms. Race is guessed from a surname and a neighbourhood. And in the redlining chapter, eighty years of housing history is joined to the census by asking which 1930s map shape happens to contain a single point.

A category rewritten while the ground stays still. The census's racial categories are not facts about people. They are a federal classification, rewritten in 1997 and again in 2024, and a trend line that crosses one of those rewrites is partly measuring the rewrite rather than the country.

Two chapters take this apart. One shows a single population using both of the form's escape hatches at once. The other shows a sharp legal line: language coverage starts at ten thousand speakers. That line is drawn by a survey whose numbers could be off by more than the gap between a covered county and an uncovered one.

Missing values that are the finding rather than the defect. In the election-administration chapter, jurisdictions report rejections without reporting the returns those rejections came from. So every national rate is a floor, and the bias runs one way. In the lobbying chapter, a filing reports one dollar figure against a list of issues, with no split between them. So how much was spent lobbying on health care cannot be recovered from any number of filings.

In the traffic-stop file, whether contraband was found exists only for drivers who were searched. So the comparison case is not merely absent from the data. It was never recorded by anybody, anywhere.

110 of the 116 chapters carry a late section on what their evidence will and will not bear, and 116 finish by listing their sources. Those are not disclaimers. They are the part of the argument that tells you how far you may take it, and the part most worth copying.

08 What this book is not

It is **not a statistics text**. There is arithmetic throughout, and occasionally a simulation or a margin of error: how far off an estimate could be and still be doing its job. But no chapter's purpose is to teach a procedure, and several exist to show a well-regarded procedure producing a confident wrong answer.

It is **not a programming manual**. The code is base R, it is complete, it is on the page, and you are never asked to write any. If you want to learn R, this is not the book, though reading these chapters will not hurt.

It is **not a survey of American politics**. The topics are chosen for what they reveal about evidence, not for coverage. Whole subjects that belong in any politics course are absent here because no dataset makes their difficulties visible, and some minor-looking subjects get a chapter because one does.

And it does not claim that data settles arguments. Most of these chapters end with a question still open – and say which one, and say what evidence would be needed, and say why this file cannot supply it.

09 Sources

Every figure quoted above comes from the chapter it belongs to, read out of that chapter's own tables.

- The 11001 row, the 3,104 matched counties, the median swing, and the 328 codes broken by a numeric read: the election-returns chapter's 2020 and 2024 county files.
- The two traffic-stop denominators: the policing chapter's stop counts and its American Community Survey denominators, for San Francisco.
- The New Jersey registration total and the 1 January placeholders: the false-matches chapter's `data/facts.csv`, whose `source` column carries the citation for each.
- The 37.2-point step across Detroit's northern boundary: the demographics chapter, from the 2020 Census at block level.
- The counts of chapters, of chapters with a data directory, of prediction prompts and of closing limits sections were obtained by opening the 116 chapter files in the folder above this one and counting.